

3rd Set of Problems

1. Recalling that the order of transformations is defined, according to Denavit-Hartenberg's convention, through a translation d_i along z_{i-1} followed by a rotation ϑ_i about z_{i-1} followed by a translation a_i along x'_i followed by a rotation α_i about x'_i , where frame i' is an intermediate frame between the original frame $i - 1$ and the transformed frame i . Illustrate the aforementioned sequence of D-H transformations.

2. Consider the manipulators illustrated in Figure 1. According to the Denavit-Hartenberg methodology, place the reference frames over each manipulator, and build the appropriate table of parameters.

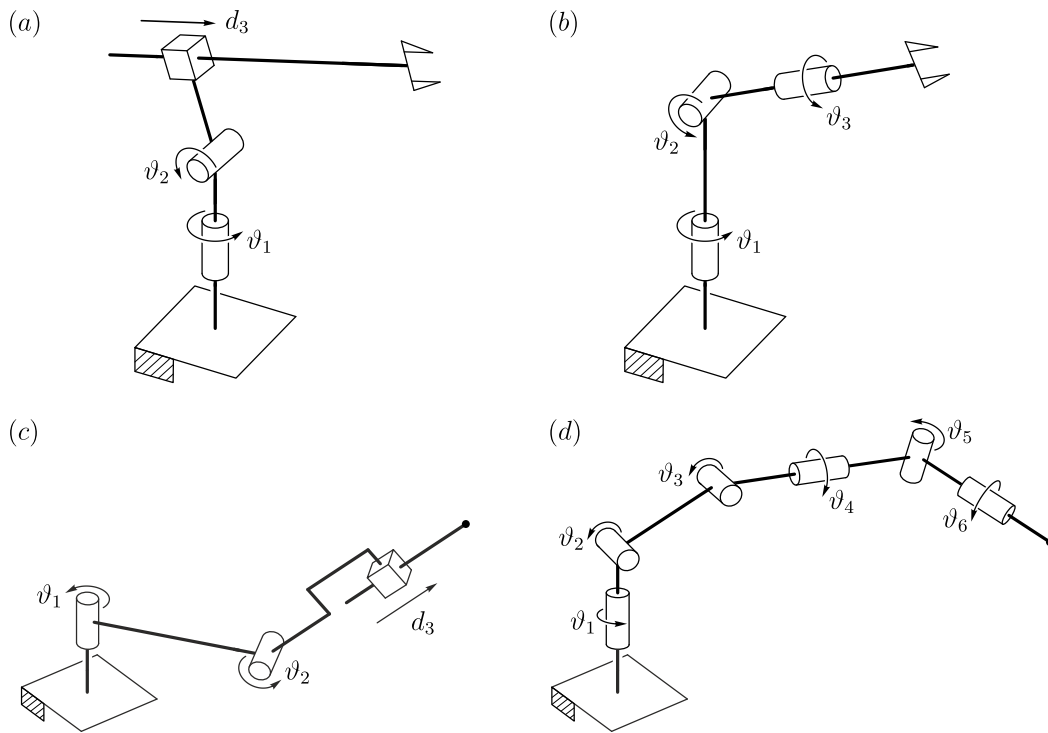


Figure 1: Manipulators. (a) Spherical, (b) Spherical wrist, (c) Toroidal, (d) Anthropomorphic.

3. Determine the spherical wrist's orientation (Figure 1b). Write an equivalent sequence of Euler angles, as a function of the manipulator's joint angles.

4. Solve the direct kinematics for the manipulator illustrated in Figure 1(c).

5. Determine the position of the anthropomorphic arm's wrist (Figure 1d), where the wrist is the point of intersection between axes z_5 and z_4 .

6. Consider the Spherical manipulator's configurations illustrated in Figure 2. For each configuration, choose the offset parameters $\sigma = (\sigma_1, \sigma_2, \sigma_3)^\top$, in the D-H Table 1, in such a way that a null joint configuration, $\mathbf{q}' = (\vartheta'_1, \vartheta'_2, d'_3)^\top \equiv \mathbf{0}$, corresponds to the illustration. Note that $\mathbf{q} = \mathbf{q}' + \sigma$.

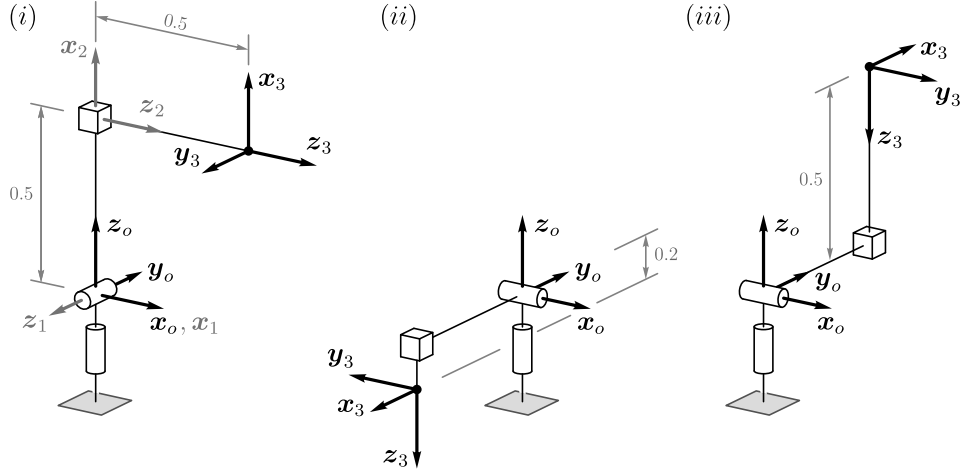


Figure 2: Spherical Manipulator's configurations.

Table 1: Spherical manipulator's D-H parameters.

Link	d_i	ϑ_i	a_i	α_i
1	0	$\vartheta'_1 + \sigma_1$	0	$\pi/2$
2	0	$\vartheta'_2 + \sigma_2$	0.5	$\pi/2$
3	$d'_3 + \sigma_3$	0	0	0

7. Given the homogeneous transformation $\mathbf{A}_e^b$, determine the inverse transformation, $\mathbf{A}_b^e$.

$$\mathbf{A}_e^b = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & c_\theta & -s_\theta & 2 \\ 0 & s_\theta & c_\theta & 3 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

8. For a given application, suppose that the orientation of the manipulator's end-effector is known with respect to the base, through

$$\mathbf{R}_e^b = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{pmatrix} = \begin{pmatrix} 0.2493 & -0.4569 & 0.8539 \\ -0.9571 & -0.2507 & 0.1453 \\ 0.1476 & -0.8535 & -0.4997 \end{pmatrix}.$$

- Express the rotation as an angle and axis rotation, to verify that $\theta \approx 138.6^\circ$ and $\mathbf{r} \approx (-0.76, 0.53, -0.38)^\top$.
- Express the rotation as a quaternion, to verify that the scalar part is $\eta \approx 0.35$ and the vector part $\boldsymbol{\varepsilon} \approx (-0.71, 0.50, -0.35)^\top$.